

Technical Feasibility & Concept Validation

Refining urban startup concepts before architecture, technology selection and municipal integration.

Urban Up V2 – Techno-Park Palestine

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01

**Validate the concept
before you build it.**

Four sessions. One build sequence.

Each session should leave you with a decision-ready output—not just more slides.

1

Technical Feasibility & Concept Validation

Can this concept work in the real municipal context?

OUTPUT: validated concept + evidence gaps

2

Prototype Architecture & Technical Requirements

What exactly must the prototype contain and how will it work?

OUTPUT: architecture + requirements

3

Technology Stack & Integration Design

Which technologies fit—and what must they integrate with?

OUTPUT: stack + integration map

4

Municipal Workflow, User Journey & Data Requirements

How does the solution fit the actual municipal process and data reality?

OUTPUT: future workflow + data map

What I expect you to bring to all four sessions

These are refinement sessions. Arrive with evidence, assumptions and open questions—not a polished story.

01

Bring the real municipal conversation

Names/roles of people you spoke to, what they challenged, what they would actually use.

02

Bring evidence—not only enthusiasm

Screens, prototype, data, field observations, quotations, cost assumptions, failed tests.

03

Separate facts from assumptions

A confident assumption is still an assumption until someone or something validates it.

04

Be willing to narrow

A smaller concept that works beats an integrated platform that cannot be implemented.

Think About

What is the one thing the municipality told you that forced you to change your idea?

Technical Feasibility ≠ “Can we code it?”

Feasibility is whether the complete solution can work, be operated, maintained and justified under real constraints.

TECHNICAL

Can the core function perform reliably?

OPERATIONAL

Can it survive the real workflow and environment?

INSTITUTIONAL

Can the municipality own, approve and operate it?

ECONOMIC

Can the value justify lifecycle cost?

EVIDENCE

What proves each critical assumption?

A concept is feasible only when the critical assumptions across these dimensions are acceptable—or have a credible test plan.

Think About

Which of these five dimensions is currently your weakest?

Turn the concept into an assumption map

The fastest way to improve a concept is to expose what must be true for it to succeed.

MUST BE TRUE Example	Evidence today What do we actually know?	Failure consequence What happens if wrong?	Next test Cheapest decisive test
Sensor data is accurate enough	Lab test only	Wrong municipal decisions	Co-locate against reference
Retrofit fits existing assets	Measured 2 poles	Installation fails at scale	Survey 20 representative assets
Municipal staff will use dashboard	Positive meeting	Tool becomes unused	Observe current workflow + task test

Think About

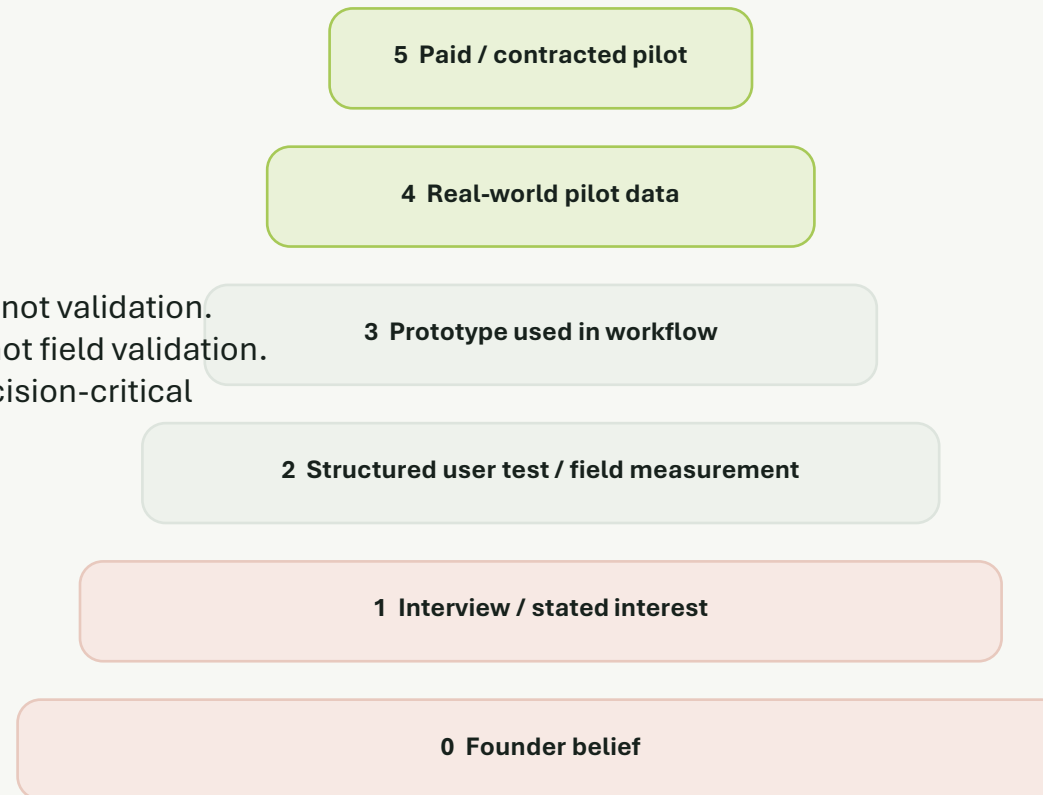
Which assumption, if false, kills your startup? That is the one to test first.

Not all validation evidence is equal

Move from opinions toward behaviour and real-world performance.

VALIDATION LADDER

- A municipality saying “interesting” is not validation.
- A prototype working on your desk is not field validation.
- A pilot is valuable only if it tests a decision-critical assumption.



Think About

What is the strongest evidence you currently have—not the strongest claim?

Validating the solution instead of the failure

Municipal teams often agree that a technology is “useful”. That does not prove the problem is painful enough to adopt it.

WEAK

“Would a smart bin / dashboard / AI tool be useful?”

BETTER

“Show me the last time this process failed. What happened next? What did it cost?”

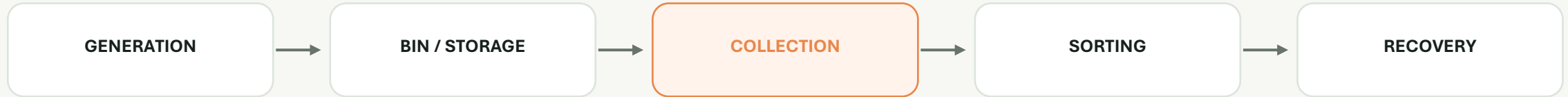
Lesson learned: validate the current failure, frequency, workaround and consequence before validating your proposed feature.

Think About

Can you describe the last real incident your solution would have prevented?

Waste: test the bottleneck—not the buzzwords

A “smart waste solution” may fail because the wrong assumption was tested.



Claim

“Fill sensors will reduce collection cost.”

Hidden assumptions

- Bins are the routing constraint
- Routes can change operationally
- Connectivity is reliable

Decisive pilot

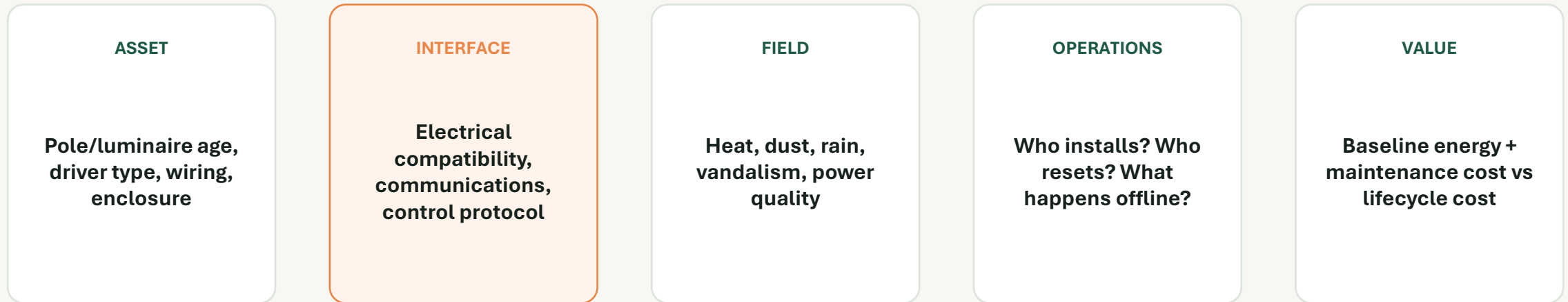
**Compare baseline vs. dynamic routing:
km/tonne, overflow, hours, fuel.**

Think About

If the sensor works perfectly but routes cannot change, is the concept feasible?

Street lighting: retrofit feasibility is mostly interfaces

The algorithm may be easy. Existing assets, installation conditions and maintenance reality are often the hard part.



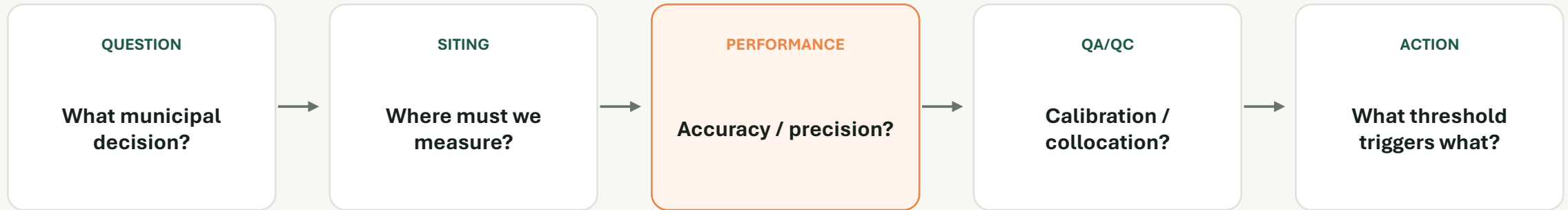
Pilot lesson: survey representative assets before freezing the product. One compatible pole proves almost nothing about a citywide retrofit.

Think About

What percentage of the municipality's existing assets can your concept actually work with today?

Environmental sensing: “it produces data” is not enough

Low-cost sensing is useful, but accuracy, siting, calibration, maintenance and intended use determine whether the data is decision-grade.



EPA lesson: plan the monitoring question first; evaluate sensor performance and data quality for the intended use. Low-cost sensors are not automatically equivalent to regulatory monitors.

Think About

What decision would you refuse to make using your current data?

Municipal digital platforms: feasibility lives behind the screen

The UI can work while the service fails because approvals, data ownership and existing systems were never validated.

Citizen / field user

Submit / report / request

Municipal officer

Verify → classify → act

Manager

Approve / prioritise / escalate

Existing systems

Registry / finance / GIS / identity

FEASIBILITY
CHECK

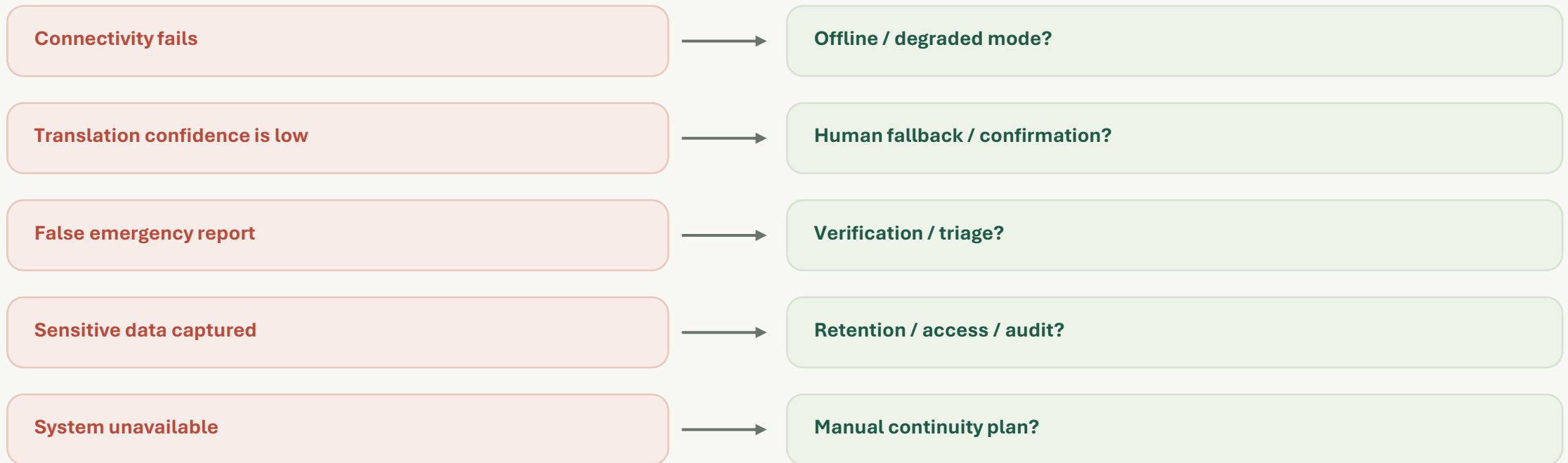
- Who owns the record?
- Can data be exchanged?
- What happens when data conflict?
- Can staff bypass the new workflow?

Think About

If the municipality cannot give you API access, what is your viable fallback?

Accessibility & emergency tools: design the failure mode first

For critical services, feasibility includes what happens when the ideal conditions disappear.



A safety-critical concept is not validated by the happy path. Validate the degraded path.

Think About

What happens at 2 a.m. when your system is wrong, offline or ambiguous?

The “platform gravity” trap

Teams keep adding modules until the concept becomes impossible to validate.



Better question: What is the smallest end-to-end capability that proves the municipal value?

Think About

What can you delete from your concept without weakening the core value proposition?

Pilot theatre: a pilot that cannot answer a decision

“We installed it and people liked it” is not a validation result.

PILOT THEATRE

- No baseline
- No acceptance threshold
- Too many variables change
- Only friendly users
- No owner for post-pilot decision

DECISION PILOT

- Baseline before intervention
- Pre-agreed success / failure threshold
- Test one critical assumption
- Representative users/assets
- Define: scale / revise / stop

Think About

What exact decision should the municipality be able to make at the end of your pilot?

GovTech practice: pilots should generate learning that informs scaling and future procurement.

https://www.oecd.org/en/publications/digital-government-review-of-biscay-spain_ab35c0ea-en/full-report/assessment-and-recommendations_4c0604a0.html

The Feasibility Gate: evidence before architecture

Before Session 2, each team should be able to pass—or explicitly flag—these gates.

G1

Problem evidence

Repeated + consequential

G2

Core function

Technically plausible

G3

Field conditions

Constraints understood

G4

Municipal ownership

User + decision owner named

G5

Lifecycle

Install / operate / maintain credible

G6

Value

KPI + baseline identified

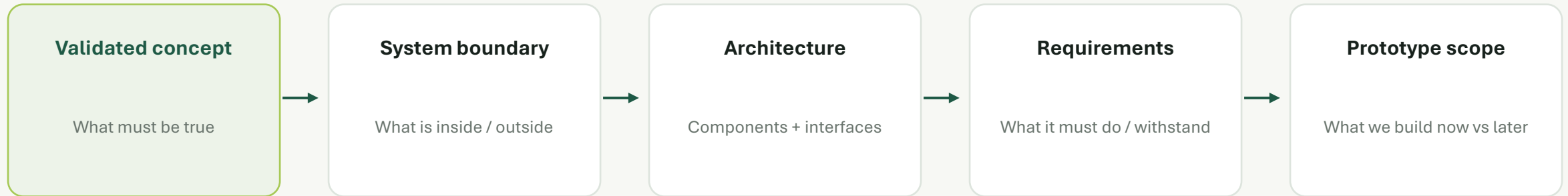
RED does not mean “bad startup”. It means “do not freeze the architecture yet”.

Think About

Which gate would a skeptical municipal engineer mark RED today?

From a validated concept to a buildable prototype

Session 2 starts only after you know what must be proven. Then we decide what the prototype must contain.



Think before next time: What is the system boundary? What are the 3–5 core functions? What external systems/assets must your prototype interface with?

Think About

If you could prototype only one end-to-end workflow, which one would prove the most?

The Feasibility Red-Team Sprint — 60 minutes

Do not improve the pitch. Try to break the concept, then redesign the validation plan around the weakest assumption.

01

10 min

Define one municipal use case

One user + one job + one measurable failure.

02

15 min

List 8 assumptions

Technical, field, institutional, lifecycle, value.

03

10 min

Rank the kill assumptions

Score: uncertainty × consequence. Select top 2.

04

15 min

Design decisive tests

Evidence, test method, threshold, owner, time/cost.

05

10 min

Municipal challenge

Another team attacks the assumptions; revise one.

Leave with a one-page Concept Validation Record

This becomes the handover into Session 2.

A

Municipal use case

Who does what, where, and why it matters

B

Concept boundary

What the solution does—and deliberately does not do

C

Top 2 kill assumptions

The assumptions most likely to invalidate the concept

D

Evidence today

What is proven vs inferred vs unknown

E

Decisive validation tests

Method + metric + pass/fail threshold

F

Architecture implications

What must remain flexible until the tests are complete

SUCCESS = you discover what you do not yet know, and know exactly how you will test it.

Do not defend your concept. Try to disprove it with Minimal Cost.

If it survives the right tests, then it deserves an architecture.

What must be true?

What evidence do I have?

What could kill it?

What is the cheapest decisive test?